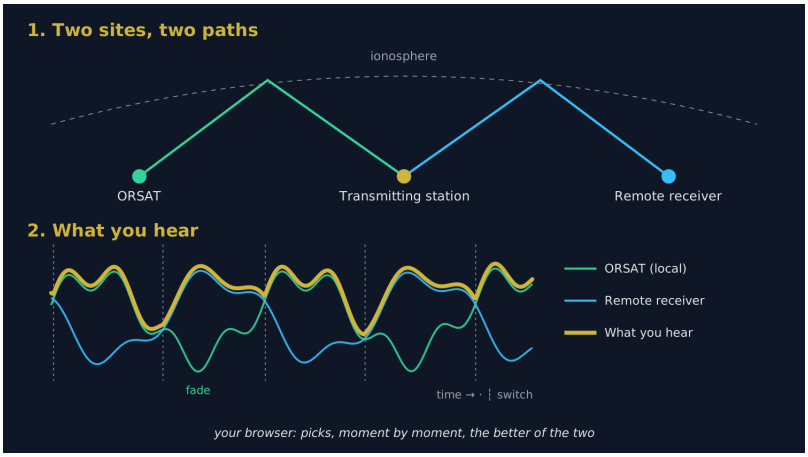
Receive diversity

Detailed guide — the principle, choosing a second receiver, reading the indicators
https://orsat.ddns.net



Two distant receivers do not hear the same fades: ORSAT switches, moment by moment, to the better of the two.

This guide explains why a shortwave signal fades, how a second receiver located elsewhere lets you ride through those fades, and how to use ORSAT's “Receive Diversity” panel step by step.

English version — October 2026 edition — ORSAT 1.0
Receive diversity developed by **SV1BTL** for PhantomSDR-Plus, integrated and documented by **F1NSK** for ORSAT.

The screenshots show the interface in French; the button names in the text are those of the English interface.

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1. The problem: QSB

1.1 Why a signal fades

On shortwave, the signal of a distant station does not reach you in a straight line: it bounces off the **ionosphere**, a layer of the atmosphere that never stops moving. It often arrives by **several slightly different paths at once**. Those paths add up, sometimes reinforcing each other, sometimes cancelling out.

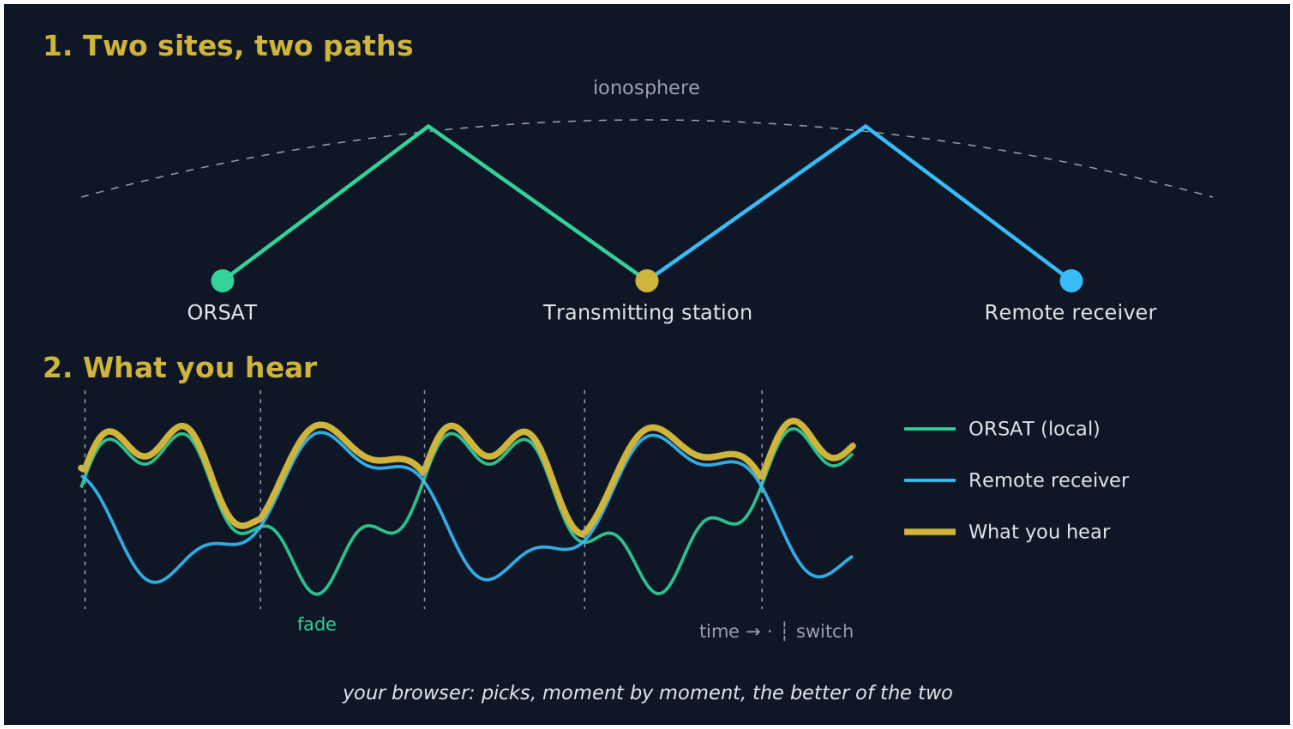
The result: the signal **rises and falls** every few seconds. This is **QSB**, or fading. At the bottom of a fade the station can vanish completely for a second or two, just long enough to lose a callsign or a report.

1.2 What a single receiver cannot do

Turning up the volume, changing the filter or switching on noise reduction changes nothing: when the signal cancels out **where the antenna is**, there is nothing left to amplify. The only remedy is to listen **somewhere else**.

2. The solution: listen in two places

2.1 The principle



Two distant receivers hear the same station over two different paths. Their fades do not happen at the same moment: ORSAT follows, moment by moment, whichever of the two receives better.

Two receivers several hundred kilometres apart hear the same station over **two different ionospheric paths**. Their fades are **independent**: when the signal dips at ORSAT, it is usually fine at the other receiver, and the other way round.

Receive diversity connects ORSAT to **a second web receiver located elsewhere**, on the **same frequency**, and lets you hear **the better of the two** at every moment. The dips of one are filled in by the other: the station stays readable where, with a single receiver, it would have disappeared.

> ***Tip :*** *The gain is measured in* ***continuity*** *rather than strength: diversity does not make the signal louder, it stops it from disappearing.*

2.2 Why choose rather than add

You might think adding the two sounds together would be even better. In fact the two paths have different phase and a slightly different frequency offset: added together, the two sounds give a **hollow, metallic** voice, as if spoken down a pipe. Clean adding would need two receivers locked to the same clock, which is impossible between two sites linked over the internet.

ORSAT therefore **chooses** between the two (this is called *selection combining*). When it switches from one to the other, it makes a **30-millisecond crossfade**, too short to be heard: there is no audible break.

2.3 How ORSAT decides

- It continuously measures the **signal-to-noise ratio (SNR)** of both receivers.
- It only switches if the other receiver is **better by at least 2.5 dB**, and only if the current one has held for **at least 0.4 second**. This avoids jumping back and forth on every syllable.
- If the current receiver **collapses** (below 6 dB for half a second), it switches without waiting.
- If the remote receiver **stops responding** for 1.5 seconds, ORSAT falls straight back on itself. **Diversity never cuts the sound**: at worst you hear ORSAT alone, as usual.

3. The delay between the two receivers

3.1 Why alignment is needed

The remote receiver's audio does not reach you at the same moment as ORSAT's: each receiver has its own buffers, and the internet path adds its own delay. The gap is usually between a few tenths of a second and 1.5 seconds, and can be up to 4 seconds.

Without correction, switching from one receiver to the other would **skip or repeat** part of a sentence. ORSAT must therefore first measure that exact delay.

3.2 How it is measured

ORSAT compares the **rhythm of speech** in the two sounds: the sequence of syllables and pauses, which stays recognisable despite the different paths and audio compression. It looks for the offset that makes the two rhythms line up.

- It first searches a **±1.5-second** window (about 9 seconds of listening), which is nearly always enough, then widens to **±4 seconds** if needed.
- It waits for **two successive measurements to agree** within 40 milliseconds before declaring itself aligned.
- Once aligned, it **re-checks the delay every minute**, since the clocks of the two sites drift very slowly.

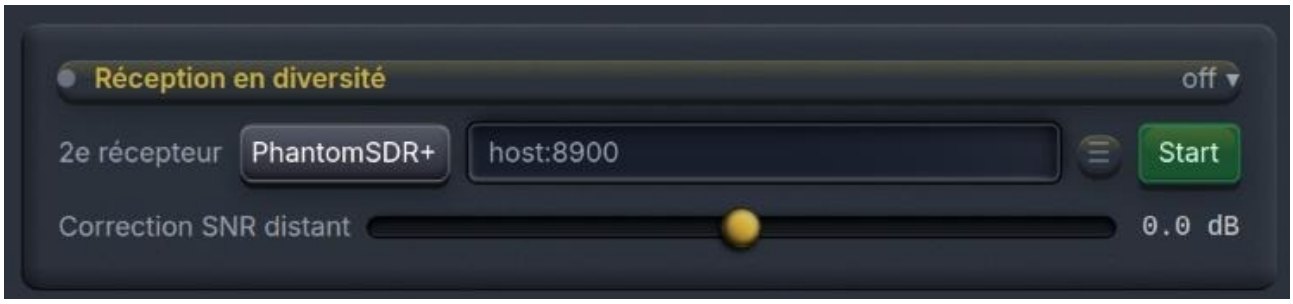
Allow **10 to 20 seconds** between pressing **Start** and diversity being operational. **Someone must be talking** during that time: on a bare carrier or an empty frequency there is no rhythm to compare.

> ***Tip :*** *Once aligned, the sound you hear is about* ***0.3 second behind*** *ORSAT alone. That is the price of the margin that allows smooth switching; when listening, you do not notice it.*

4. Using the panel step by step

4.1 The panel

The **Receive Diversity** panel is in the centre column, below fine tuning. Click its title bar to unfold it.



The “Receive Diversity” panel unfolded.

4.2 Starting

1. **Tune ORSAT first** to the station you want, in the right mode.
2. In **Second RX**, choose the type of the remote receiver: **PhantomSDR+**, **KiwiSDR**, **UberSDR** or **WebSDR** (PA3FWM's classic WebSDR).
3. In the address field, type its address as *host:port* (for example **mywebsdr.example.org:8073**). The greyed text in the field shows the expected form for the chosen type.
4. Click **Start**.
5. Wait: the title bar shows **Please wait…** during alignment, then **Ready** when diversity is operational.

You do not have to tune the remote receiver. ORSAT automatically puts it on the same frequency, mode and filter as itself, and follows every retune.

To stop, click **Stop**: you are instantly back to ORSAT alone.

4.3 The title-bar light

Colour	Meaning
Grey	Diversity off
Blinking amber	Connecting or aligning
Blinking green	Aligned, you are hearing **ORSAT**
Blinking cyan	Aligned, you are hearing **the remote receiver**

Listening to a fluctuating station, you will see the light switch between green and cyan as the fades come and go: that is diversity at work.

4.4 The detailed indicators

While running, a table shows the state of the link:

Indicator	What it tells you
link	State of the connection to the remote receiver (connecting, ready, closed, error) and the reason for a drop
aligned	Measured delay between the two sounds, in seconds; “searching” until it is found
corr	How alike the two sounds are once aligned: the higher, the more reliable the alignment
remote audio	Amount of audio received from the remote and its sample rate; “none” means it is sending nothing
SNR local / SNR remote	Signal-to-noise ratio of each receiver, in dB, on which the choice is made
switches	Number of changes from one receiver to the other since starting

If the message **“remote does not cover this frequency”** appears, the second receiver does not receive the band you are on: choose another one.

4.5 Remote SNR trim

The **Remote SNR trim** slider (−15 to +15 dB) is added to the remote receiver's SNR before comparison.

- With **another PhantomSDR+**, leave it at **0 dB**: the two measurements are directly comparable.
- With a receiver that **compresses its audio heavily** (a classic WebSDR sends 8-bit audio), its noise floor is partly hidden by the compression and its SNR looks lower than it really is. Raise the trim by a few dB if ORSAT never hands over to it although it is clearly better.
- Conversely, lower it if ORSAT switches too readily to a remote receiver that sounds worse.

5. Saved receivers

The ☰ button, next to the address, opens your list of receivers, kept **per type**: a KiwiSDR address is only offered when the KiwiSDR type is selected.

- Every receiver started successfully is **added automatically** at the top of the list.
- **add** adds a receiver by hand; ✎ edits its name and address; × removes it; **clear all** empties the list for the current type.
- Clicking a row puts its address in the field; ▲ ▼ or drag-and-drop change the order; opens the receiver's page in a new tab.
- The receiver in use is shown in bold in the list.
- **export** saves all the lists to a file; **import** reads them back, offering to **merge** with the current list or **replace** it.
- ▦ **QR** shows the lists as a QR code: scan it with a phone to get your receivers on ORSAT's mobile page (Div tab).

These lists are kept **in your browser**: they do not travel from one computer to another without an export or the QR code.

6. Choosing the second receiver well

This is the most important choice: diversity is only worth the difference between the two sites.

- **Far enough from ORSAT**: from **a few hundred kilometres**, fades become independent. Two receivers in the same town fade together and add almost nothing.
- **Not too far either**: it must **hear the station**. For a European station, a receiver in another European country works better than one in Australia.
- **Covering the band** you are on, with a decent antenna.
- **Preferably a PhantomSDR+**: same audio quality, same SNR measurement, no trim to set.
- **Available**: a receiver overloaded with listeners or with an unstable connection will often make diversity fall back to ORSAT alone.

> ***Tip :*** *WebSDR directories such as* ***sdr-list.xyz*** *list PhantomSDR+, KiwiSDR and WebSDR receivers by country.*

7. What diversity does not do

- **It only feeds the speaker.** The decoders (FT8, CW, RTTY…) keep using **ORSAT alone**: a change of receiver mid-slot would break decoding that needs continuous audio.
- **It does not replace a good antenna.** If neither receiver hears the station, diversity cannot help.
- **It does not mix the two sounds**, except during the 30 ms crossfades (see section 2.2).
- **It cannot align on an unmodulated signal**: it needs a voice, Morse or any signal that varies over time.

8. Troubleshooting

Symptom	Likely cause	Remedy
Stays on “Please wait…”	Nobody is talking, or the remote does not hear the station	Wait for a transmission; check on the remote receiver's page () that it hears the station
link: error or closed	Wrong address, wrong type, receiver offline or full	Check the address and type; try another receiver
remote audio: none	The remote accepts the connection but sends no audio	Try another receiver of the same type
“does not cover this frequency”	The second receiver does not receive this band	Choose a receiver that covers the band
Light stays green, 0 switches	ORSAT is always better, or the remote is underrated	Normal if ORSAT is better; otherwise raise the remote SNR trim
Constant switching	The two receivers are about equal	Harmless to listen to; lower the trim slightly to favour ORSAT
A word repeated or skipped at a switch	Alignment being corrected	Fixes itself at the next check (within a minute)

Tip : *When in doubt, click* ***Stop****: the sound returns immediately to ORSAT alone, with no side effects.*